

ACPMU-RF Project Charter v0.1

1. Project Overview

Project Name: Avionics Communication, Power Monitoring & RF Telemetry Laboratory Demonstrator

2. Project Purpose

The purpose of the ACPMU-RF project is to design a modular avionics electronic system that combines power management and monitoring, wired communication, and RF telemetry functions, and to gain practical engineering experience in the system design, analysis, simulation, and verification processes.

3. Project Objectives

The project aims to:

- Design an STM32G474RE-based modular avionics board,
- Design a nominal 9-24 V DC input system and use the necessary protection, filtering, and voltage regulation applications,
- Use CAN-FD, RS-232, and RS-485 communication interfaces,
- Measure and monitor current, voltage, and power values using INA238,
- Store the measured values using microSD,
- Design a separate 2.4 GHz RF telemetry daughterboard from the main board,
- Design a 50 Ω RF transmission line, matching/filter network, and SMA antenna interface,
- Design, simulate, and optimize a 2.4 GHz antenna using CST,
- Perform electrical, thermal, SI, and RF verification simulations,
- Document system requirements, ICD, RTM, risk analyses, test results, and simulation results.

4. In Scope

The project includes:

- Definition of system requirements
- System block design
- Power system architecture and power budget
- MCU selection and pin allocation
- Clock/reset/debug architecture
- CAN-FD communication interface
- RS-232 communication interface
- RS-485 communication interface
- Current, voltage, and power measurement with INA238
- microSD interface
- RF transceiver integration
- RF transmission line and matching/filter design
- 50 Ω RF PCB trace design
- Antenna design
- Link budget calculation
- Implementation of LTspice simulations

- Implementation of CST electromagnetic simulations
- Signal integrity analysis
- Thermal analysis
- ERC and DRC checks
- PCB design and preparation of manufacturing outputs
- Interface Control Document (ICD)
- Requirements Traceability Matrix (RTM)
- Design Risk Register
- Verification plan
- Preparation of test steps
- ECSS/MIL/CCSDS applicability and awareness analysis
- Portfolio documentation

5. Out of Scope

The project does not include:

- Development of a flight control system
- Development of a guidance and navigation system
- Autonomous targeting
- Weapon systems or weapon integration
- Development of flight-ready/flight-qualified hardware
- Formal ECSS or MIL-STD certification
- Formal CCSDS compliance certification
- Space qualification
- Real mission/satellite operation
- Mandatory PCB manufacturing and physical prototype
- Mandatory VNA, spectrum analyzer, or open-field RF measurement

Results without physical testing will be stated as simulation/analysis evidence; they will not be considered physically verified.

6. Major Deliverables

The main outputs of the project:

- Project Charter
- System Requirements
- System Architecture
- Interface Control Document (ICD)
- Requirements Traceability Matrix (RTM)
- Design Risk Register
- Main Board schematic and PCB
- RF Daughterboard schematic and PCB
- Power Budget
- Thermal Analysis
- Link Budget

- LTspice Simulation Reports
- CST Antenna Design & Simulation Report
- SI Verification Evidence
- ERC/DRC Reports
- Verification Plan & Evidence
- Manufacturing Package
- Final Project README / Portfolio Documentation

7. Assumptions & Constraints

Assumptions

- The project can be completed without physical PCB manufacturing.
- Required technical parameters will be obtained from manufacturer datasheets.
- Manufacturer reference designs will be used as the basis for RF matching/filter design.
- Unknown values will be marked as TBD rather than estimated.

Constraints

- Access to physical laboratory equipment is not guaranteed.
- Without actual testing, the terms “hardware validated” or “bench tested” will not be used.
- Formal aviation certification and space qualification are outside the scope of this project.

8. Verification Strategy

The project’s primary verification approach will be simulation-first verification.

Methods to be used:

- Analysis
- Simulation
- Inspection
- Design Review
- ERC/DRC
- LTspice
- CST
- Signal Integrity Analysis
- Thermal Analysis

Where necessary, the following statuses will be used in verification outputs:

SIMULATED - ANALYSED - REVIEWED - INSPECTED - NOT EXECUTED

If physical testing is performed, it may additionally be indicated as TESTED.

9. Project Risks

Main risks to be monitored at the initial level:

- Incorrect component selection
- Insufficient power/thermal margin
- MCU pin or peripheral conflicts
- Logic-level incompatibilities
- CAN/RS-485 termination errors

- PCB ground/return path issues
- Signal Integrity issues
- RF impedance mismatch
- Antenna detuning
- Incorrect footprint
- Component availability
- Deviation of simulation models from the actual system

Detailed evaluation will be maintained in a separate Design Risk Register.

10. High-Level Schedule & Milestones

Week	Main Focus	Main Output
1	Requirements & System Architecture	Project Charter, SYS-REQ, RTM
2	STM32 & MCU Hardware	MCU schematic, pin/clock plan
3	Power Electronics	Power design, LTspice, thermal budget
4	Communication Interfaces	CAN-FD, RS-485, RS-232, ICD
5	RF System Design	RF architecture, link budget
6	Antenna & RF PCB	CST antenna, RF layout
7	Full Schematic & BOM	Full schematic, ERC, BOM
8	PCB Design	Routed PCB, DRC, fab outputs
9	SI / Thermal / Standards	Verification & standards evidence
10	Final Verification	Final RTM, reports, portfolio

11. Success / Completion Criteria

The project will be considered successfully completed under the following conditions:

- System requirements have been defined
- System architecture has been completed
- The main board and RF daughterboard have been designed
- Power and communication interfaces have been documented
- PCB ERC/DRC checks have been completed
- LTspice and CST analyses have been performed
- Power, RF, thermal, and SI verification outputs have been created
- Requirements have been linked to verification evidence through the RTM
- Risks and open issues have been documented
- Manufacturing and portfolio outputs have been prepared

12. Tools & Standards

Engineering Tools

- Altium Designer
- STM32CubeMX / CubeIDE
- LTspice

- CST Studio Suite
- Git
- Spreadsheet / Calculation Tools

Standards Awareness

- ECSS
- MIL-STD
- CCSDS

These standards will be evaluated only at the applicability / standards awareness / pre-compliance level; no formal compliance or certification claim will be made.